

Cultural traits can evolve. Here we use a model of vertical cultural transmission developed by Cavalli-Sforza & Feldman 1981, where a dichotomous cultural trait (with types  $b$  and  $B$ ) is passed from parents to offspring. The cultural type of the offspring is determined by a set of transmission parameters,  $g_i$ , as shown in the following table.

| Mating type |          | Fraction of   |               |
|-------------|----------|---------------|---------------|
| parent 1    | parent 2 | $B$ offspring | $b$ offspring |
| $b$         | $b$      | 0             | 1             |
| $b$         | $B$      | $g_1$         | $1 - g_1$     |
| $B$         | $b$      | $g_2$         | $1 - g_2$     |
| $B$         | $B$      | 1             | 0             |

Let  $x(t)$  be the frequency of the cultural innovation,  $B$ , in generation  $t$ . Assuming random mating, the frequency of the cultural innovation in the next generation is then

$$x(t+1) = x(t)(1-x(t))g_1 + x(t)(1-x(t))g_2 + x(t)^2.$$

1. [6/20 points] Show how to find the two equilibria,  $\hat{x} = 0, 1$ .
2. [4/20 points] Given  $g_1$  and  $g_2$  are fractions, what are the minimum and maximum values  $g_1 + g_2$  can take?
3. [8/20 points] Show how to determine that  $\hat{x} = 0$  is only stable when  $g_1 + g_2 < 1$  and  $\hat{x} = 1$  is only stable when  $g_1 + g_2 > 1$ .
4. [2/20 points] Which equilibrium is stable in the plot below?

